



RECOVERY ARCHITECTURE

SkyRipple

Solving the Multiplier Effect

Agentic AI and discrete-event simulation engine for multi-day airline network recovery.

NETWORK CASCADE

The Multiplier Blind Spot

A local disruption cascades into massive network-wide financial losses.

- › Direct cost of a 2-hour runway reduction at ORD is **\$181K**.
- › This disruption cascades into a severe **\$942K** network loss.
- › The result is a **5.19x multiplier** effect that traditional tools fail to mitigate.

5.19x

The Multiplier Effect

Traditional tools fail to mitigate this compounding network cascade.

AGENTIC SIMULATION

Arbitrated Financial Simulation

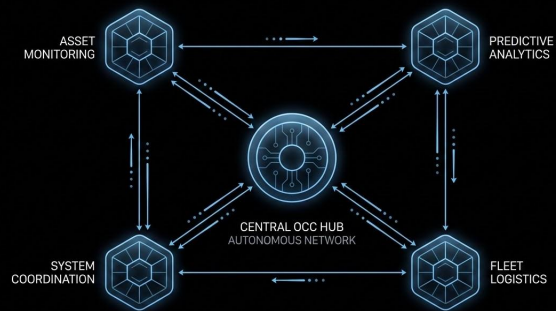
Agentic simulation outperforms traditional rule-engines in adversarial recovery.

- **5-Agent OCC:** Powered by a specialized 5-agent Operations Control Center.
- **Duty Manager:** Executes parallel financial simulations to optimize recovery paths.

ADVERSARIAL RECOVERY

\$172,854

Proven savings over traditional rule-engines in adversarial recovery tests.



AUTONOMOUS OPERATIONS CONTROL CENTER NETWORK (OCC)
DECENTRALIZED GEOMETRIC SCHEMATIC

SYSTEM ARCHITECTURE

Gemini Proposes. Code Disposes.

A strict operational boundary separating natural language reasoning from deterministic execution layers.

01 / PARSE

Gemini Proposes

Gemini parses natural language directly into strict Pydantic schemas.



02 / EXECUTE

Code Disposes

A deterministic code engine executes all mathematical and simulation logic.



03 / ARBITRATE

Zero Hallucination

This strict boundary ensures zero hallucination during critical operational arbitration.

Enterprise-Grade Verification

SkyRipple operates at massive scale with zero latency.

582,304

Grounded: Validated against 582,304 real BTS records (Dec 2025) and FAA registries.

\$5.8M

Emergent Insights: Surfaced counter-intuitive network windfalls, including a \$5.8M cost swing from competitor groundings at DEN.

8s

Memoization: Aggressive caching dropped massive multi-day passenger itinerary generation from 110s down to 8s.

The \$30B Efficiency Gap

Multi-day recovery represents a massive global opportunity.

Market Breakdown

- **TAM (Global):** \$30B+ lost annually to systemic delays across global commercial aviation and logistics.
- **SAM (US Domestic):** Major carriers operating complex, high-vulnerability hub-and-spoke models.
- **SOM (Top 5 Airlines):** The largest US operators facing immediate FAA regulatory pressure and crew-legality bottlenecks.

\$30
BILLION

Systemic Operational Efficiency Gap in Global
Airline Recovery

Deterministic Intelligence

Real-time financial simulation surpasses static rule-engines.

Legacy Rules

Mathematically reliable, but rely on blunt, static logic that breaks during multi-day cascades.

Standard LLMs

Highly adaptable and intelligent, but structurally prone to hallucinating financial outcomes.

SkyRipple

Occupies the only viable enterprise quadrant—adaptive intelligence through deterministic, safety-by-construction arbitration.

BUSINESS MODEL

Scalable licensing aligns with operational compute demands.

PLATFORM

Core OCC SaaS

Annual recurring seat & hub license for Airline Operations Control Centers.

Includes live disruption parsing, real-time cascade visualization, and standard single-day recovery.

COMPUTE & SCALE

Multi-Agent Simulation

Tiered/consumption billing for parallel continuous-month simulations and heavy agentic arbitration.

Dedicated high-throughput VPC instances for zero-latency operational recovery.

PERSPECTIVE

Engineering Meets Strategy

Bridging computer science with operational management.

- Built by **Aaryan Sharma**, Systems Engineer.
- Master in Management candidate at **Kellogg School of Management**.
- Focused on solving **multi-million-dollar** operational bottlenecks.



Aaryan Sharma

Systems Engineer

Master in Management
Kellogg School of Management

The Next Era of Ops

Partner with us to orchestrate global operational recovery.

Seeking agreements with pilot airlines for initial live deployment.

Validating technical superiority in real-world multi-day recovery environments.

Establishing the definitive path to global operations orchestration.

Our Vision

"Orchestrating Global Operational Recovery"